7. Existing Mitigation Strategies: Who Benefits from Protection?

This section evaluates existing privacy mitigations through a patient-level lens. It distinguishes formal guarantees from empirical reductions in attack performance and asks not only whether a method lowers average leakage, but also whether it protects patients who were initially most exposed, reduces subgroup disparities, preserves clinical utility, and avoids transferring risk to others.

Table 7.1. Comparison of mitigation families considered in Section 7

Mitigation family	Mechanism and protection claim	Current patient-level evidence	Principal limitation
Differential privacy	Contribution clipping, random noise, and accounting; provides a formal (ε, δ) guarantee for the declared adjacency relation and protected unit	Direct clinical evidence that stronger record-level DP can shift patient-vulnerability distributions downward [3]	Record-level adjacency does not automatically protect a patient's complete multi-record contribution
Regularization and training modifications	Early stopping, weight decay, augmentation, label smoothing, or adversarial regularization; protection is empirical and attack-dependent	Limited direct patient-level evidence; regularly trained clinical models can retain an extreme-vulnerability tail [3]	Effects depend on the attack, data, capacity, training configuration, and utility cost
Data and representation interventions	Rebalancing, contribution control, deduplication, synthetic data, or feature modification; usually no formal guarantee	Sparse clinical evidence concerning protection of the same patients before and after intervention	May change representation, redistribute exposure, or reduce rare-group utility without eliminating leakage
Distillation and output-access restrictions	Teacher-student transfer, label-only release, rounding, noise, query limits, or monitoring; reduces observable signals or operational attack capacity	Per-example non-clinical evidence shows that leakage can persist through distillation [44]	Students may inherit teacher leakage, and adaptive attacks can exploit restricted interfaces
Personalized and risk-adaptive methods	Non-uniform budgets, sampling, clipping, or risk-dependent weighting; formal only for the protected unit and accounting specified	No validated end-to-end clinical patient-level pipeline was identified	Audit error, multi-record adjacency, unequal utility effects, and risk transfer remain unresolved

7.1 Differential Privacy and the Unit of Protection

Differential privacy bounds a randomized mechanism under a declared adjacency relation [41]; attack reductions measure one adversary under one protocol. Neither failed attacks nor low empirical AUC certify DP, and a formal guarantee does not make all patients' empirical vulnerability equal. Section 3.4 sets out the distinction.

The meaning of the guarantee depends fundamentally on the adjacency relation. Under record-level differential privacy, neighboring datasets differ in one training record. Under patient-level differential privacy, they differ in the complete set $\mathcal{R}_p$ of records contributed by one patient. These definitions coincide only when every patient contributes exactly one record. Group privacy can extend a record-level guarantee to a patient contributing m_p records, but with weaker parameters. For pure ε-DP, the resulting bound can deteriorate to approximately $m_p\varepsilon$. Under approximate (ε, δ)-DP, both parameters deteriorate according to the applicable group-privacy result, with the accumulated δ also depending on group size and ε [41]. Group privacy is consequently a mathematical fallback, not evidence that a record-level mechanism offers the original privacy budget to the patient's complete contribution.

DP-SGD clips per-contribution gradients, adds calibrated Gaussian noise and composes the loss across sampled updates [8]. The guarantee depends on correct sampling, noise, steps, adjacency and accounting; clipping alone is not DP. If patients contribute multiple records, record-gradient clipping does not automatically bound a whole patient's effect.

The distinction becomes clinically consequential when one patient contributes multiple correlated records. A record-level private optimizer clips and protects each record separately; it does not necessarily bound the combined contribution of all m_p records by the same parameters. Knolle et al. found that stronger record-level DP-Adam was associated with lower patient-level membership vulnerability [3]. However, some patients contributed several records, and patient-level scores were derived from the most exposed record. Patient-level attack measurements can therefore exceed a bound interpreted for a one-record change without contradicting the underlying record-level guarantee. The study supports differential privacy as an effective mitigation, but does not demonstrate that every patient received an equivalent patient-level guarantee or that the same parameters applied to the patient's complete contribution. The protected unit must be aligned before formal and empirical results are compared.

Constructing genuine patient-level protection generally requires restructuring training around patients. One principled construction samples patients as units, processes and aggregates each patient's records into a single contribution, clips the norm of that complete contribution, and then adds calibrated noise. A maximum number of records, visits, or local training steps per patient may also be imposed to control contribution size and prevent patients with extensive histories from dominating optimization. Most importantly, the accountant must use patient-sampling probabilities and patient-adjacent datasets. McMahan et al. demonstrated the analogous principle for language modeling by clipping complete user updates and defining adjacency through removal of all examples associated with one user [42]. Their results establish feasibility of user-level protection in a non-clinical setting, but do not determine its utility or fairness consequences for heterogeneous medical populations.

Formal patient adjacency bounds worst-case distinguishability; equal empirical vulnerability or utility does not follow. Knolle et al. measured a downward shift under stronger record-level DP, while patient contributions could contain multiple records [3]. Thus formal parameters and empirical patient results must name their protected units separately. Section 8 evaluates who benefited using the common protocol in Table 4.1.

7.2 Regularization and Training Modifications

Regularization and training modifications seek to reduce privacy leakage by changing either the learning objective or the trajectory through which a model is optimized. Relevant interventions include early stopping, weight decay, label smoothing, data augmentation, and membership-adversarial regularization. Unlike differential privacy, however, these procedures do not generally impose a formal, attack-independent bound on the effect of an individual's data. A reduction in the success of a particular membership-inference attack therefore constitutes empirical mitigation rather than a general privacy guarantee. Yeom et al. showed that limiting overfitting can reduce the signal available to loss-based attacks because members and non-members become less separable by their losses [10]. Nevertheless, overfitting is not necessary for all forms of membership leakage, and good aggregate generalization cannot establish that every patient is protected.

Early stopping and weight decay are commonly expected to reduce membership leakage by preventing training examples from achieving substantially lower losses than comparable unseen examples.

Early stopping limits the number of optimization steps during which member-specific patterns can be fitted, while weight decay constrains effective model complexity. Their effects nevertheless depend on the wider training configuration. Tan et al. demonstrated that increased capacity need not worsen privacy when combined with suitable regularization and early stopping; in some settings, larger models achieved both better generalization and stronger resistance to membership inference [37]. Consequently, neither model size nor a single regularization parameter determines protection independently. Comparisons should also be utility-matched: lowering attack performance by severely degrading diagnostic accuracy does not represent a clinically satisfactory privacy improvement.

The designation regularizer should not itself be interpreted as evidence of privacy protection. Kaya and Dumitras found that label smoothing could increase membership vulnerability despite its conventional role in controlling overconfidence [38]. They also showed that augmentation at intensities normally selected for predictive utility did not consistently mitigate membership inference, whereas sufficiently strong augmentation could reduce attack performance at the cost of task utility. An intervention may compress confidence scores or modify training losses without eliminating other membership signals available to a stronger attacker. Evaluations should therefore retrain and retune the attacker after the intervention is applied rather than testing only an attack designed for the undefended model. Data augmentation is considered more fully in Section 7.3 as a data-based intervention.

Adversarial regularization targets membership leakage more directly. Nasr et al. formulated training as a minimax problem in which a membership adversary attempts to distinguish training members from non-members, while the predictive model is optimized to maintain task performance and reduce the adversary's gain [43]. Their non-clinical benchmark experiments showed that this approach could substantially lower black-box membership-inference success with a limited reduction in predictive accuracy. However, the method does not provide a differential-privacy guarantee: its protection depends on the modelled adversary, optimization procedure, and weight assigned to the privacy-utility objective. A model may learn to resist the adversary used during training while remaining vulnerable to attacks exploiting different signals. Moreover, the available evidence does not establish protection for extreme-risk patients, correlated multi-record contributions, or patient-level membership hypotheses.

The clinically important question is therefore not merely whether regularization lowers average attack accuracy, but which patients benefit. Knolle et al. trained clinical models using measures intended to support generalization, including augmentation, weight decay, learning-rate scheduling, and checkpoint selection, yet still identified a small tail of patients with very high membership vulnerability [3]. This finding demonstrates residual exposure in regularly trained clinical models, but it does not isolate the causal effect of any individual procedure. Similarly, Song and Mittal found heterogeneous changes in sample-specific privacy vulnerability under evaluated defenses: lower average estimated scores did not imply uniform improvement, and some samples could retain or acquire relatively high vulnerability [15]. Protection should therefore be assessed through paired patient-level changes rather than inferred solely from a shift in the population mean.

Early stopping, weight decay and adversarial regularization can weaken selected membership signals without formal guarantees [10,37,43]. Their patient-level effect remains conditional on model and attack. Table 4.1 requires an adapted attacker, paired patient estimates and clinical utility so a favorable mean does not stand in for protection of the exposed tail.

7.3 Data- and Representation-Based Interventions

Rebalancing, augmentation, deduplication, contribution limits, synthetic data and representation changes alter the material or signals the model learns. Most provide no formal patient guarantee. The unresolved issue is whether the same exposed patient benefits without losing rare clinical information [15].

Rebalancing may use oversampling of rare classes, undersampling of common classes, class-balanced sampling, or loss weighting. Truex et al. and Hu et al. associate class imbalance and long-tailed learning with unequal membership vulnerability, but these findings do not establish oversampling as a privacy remedy [30,31]. A crucial distinction is between the number of distinct patients in a group and the number of training exposures of their records. Repeating records from a small patient set

may improve numerical class balance while increasing those patients' contribution weights; recruiting additional independent patients may have a different effect. This remains a causal hypothesis requiring controlled testing. Moreover, subgroup estimates for small populations can be unstable, so apparent improvement should be accompanied by uncertainty intervals and appropriately powered comparisons [7].

Data augmentation expands transformations of existing observations rather than adding independent patients. Kaya and Dumitras found that augmentation intensities chosen primarily for predictive accuracy did not consistently mitigate membership inference, whereas stronger transformations could reduce attack success with a utility cost [38]. Its apparent protection may also change when the attacker is adapted to the augmented training procedure. The clinical tail in Section 4 persisted in a training pipeline using augmentation; that observation did not isolate augmentation's effect. Augmentation should therefore be treated as a configuration-dependent mitigation whose consequences may vary across modalities, transformation types, and correlations among a patient's records.

Contribution control includes removing near duplicates, capping records per patient, patient-balanced sampling, and limiting how frequently one patient appears across epochs. These interventions may reduce genuine training influence, but they may also lower measured vulnerability mechanically by decreasing the number of opportunities for an extreme maximum record score (Section 4.3). The two effects are not equivalent. Deleting records labeled high risk is similarly uncertain. The Privacy Onion Effect shows that removing the most exposed examples can reveal a new vulnerable layer among records that previously appeared less exposed [35]. Consequently, removal should be followed by retraining and a complete audit of all remaining patients rather than an evaluation restricted to deleted records or the original high-risk set.

Synthetic data can reduce direct circulation of original clinical records, but it is not automatically anonymous. Zhang et al. found membership leakage from health-data generators, especially for partly synthetic releases [53]. DOMIAS demonstrates that local overfitting in a generative model can permit inference about membership in its source data, with uncommon samples receiving particular attention [34]. For clinical deployment, auditing should cover three connected objects: the generative model, the released synthetic dataset, and any downstream clinical model trained on it. The central patient-level question is whether synthesis actually reduces leakage for rare patients or reproduces distinctive characteristics that preserve evidence of their contribution. Synthetic data should therefore be treated as a release mechanism requiring renewed privacy assessment, not as proof that privacy has already been achieved.

Representation-based interventions may use representation balancing, feature compression, reduced reliance on patient-specific features, or learning more shared features among clinically similar observations. The heterogeneous sample-specific posterior scores reported by Song and Mittal motivate this direction, while long-tailed learning results indicate that interventions can alter how minority classes are represented [15,31]. Neither body of evidence, however, validates representation modification as patient-level protection. Making an embedding less predictive of a demographic or sensitive attribute does not show that it lacks an individual patient signature, and suppressing subgroup information may also remove clinically useful signals. Direct clinical evidence that such methods protect the initially exposed patients remains limited.

Rebalancing, augmentation, deduplication, contribution control and representations alter both learning and the exposure opportunities of multi-record patients. Their causal effects require patient-level controls and independently adapted attacks (Table 4.1); no reviewed clinical study establishes selective, durable protection of the same extreme-vulnerability patients through these methods.

7.4 Distillation and Output-Access Restrictions

Distillation transfers a teacher's behavior to a student; output restrictions limit visible scores or queries. Both may weaken one attack without removing its underlying membership signal. Neither normally supplies a formal patient-level guarantee, so protected interfaces must face adapted attacks [5,44].

In conventional distillation, a teacher fT is trained on private data. Transfer inputs are then submitted to the teacher, and its soft labels or logits form a query set from which the student fS is trained.

Deploying the student creates three privacy objects: membership in the teacher dataset, membership in any sensitive transfer dataset, and information about teacher examples inherited by the student. These objects require separate hypotheses and audits. The student's lack of direct access to teacher records does not establish their privacy, while use of a sensitive transfer set introduces an additional membership surface.

Jagielski et al. demonstrate that students can retain membership information originating from their teachers [44]. Their LiRA-based attacks evaluated private-teacher, private-student, and self-distillation settings across four non-clinical datasets: CIFAR-10, WikiText-103, Purchase-100, and Texas-100. Although distillation reduced attack performance on average, many teacher examples received little or no per-example privacy benefit, and examples vulnerable in the teacher often remained vulnerable after transfer. Membership could sometimes be inferred indirectly from predictions on student-query examples rather than by querying the teacher on the target record itself. Leakage increased when teacher and student datasets were similar, especially in the presence of near duplicates. The attacks relied on prediction access, but their auxiliary assumptions differed: the end-to-end attack required knowledge of the student training set, whereas alternative attacks used different reference information. Removing direct teacher access therefore does not remove membership evidence. The study does not, however, establish persistence of patient-level vulnerability, multi-record exposure, or the same extreme-risk patients in clinical models.

Output-access restrictions modify the inference interface. Label-only access hides confidence scores; top-k release and rounding reduce their resolution; output perturbation adds noise; query limits cap observations; and monitoring attempts to detect abnormal patterns. Each changes the adversary's information or operational budget, but none necessarily changes what the model learned. Noise or rounding may defeat an attack calibrated to unmodified outputs while remaining vulnerable to an attacker retrained against the protected interface [15]. Query limits and monitoring may raise exploitation cost without lowering latent patient vulnerability, and their effectiveness depends on whether requests can be linked across accounts, devices, or time.

Hiding confidence values is insufficient evidence that membership leakage has disappeared. Choquette-Choo et al. showed that label-only adversaries can exploit predicted-label stability under perturbations or estimate proximity to the decision boundary, using robustness as a membership signal [5]. Confidence masking therefore removes one observable channel but may leave membership-dependent decision behavior intact. Three claims must be reported separately: eliminating leakage from the learned model, concealing the signal used by a previous attack, and increasing the query cost of exploitation. Only the first concerns removal of the underlying membership signal, and even that conclusion requires evaluation against multiple adaptive attacks.

Restrictions can also impose clinically meaningful utility losses. Confidence scores may be needed to assess calibration, communicate uncertainty, set triage thresholds, or rank patients for follow-up. Rounding or perturbation may change sensitivity and specificity at operational thresholds even when discrimination remains similar. These effects should be measured overall and within clinical and demographic subgroups, particularly for rare conditions. The clinical tail in Section 4 motivates distribution-wide audits but provides no direct evidence for these two defenses.

Distillation requires separate membership games for teacher data and any sensitive transfer data; restricted interfaces require an attacker adapted to the released interface [5,44]. Query budgets and clinical costs of suppressing scores should accompany the patient-level protocol in Table 4.1. Lower success of an old attack alone is insufficient evidence of reduced learned leakage.

7.5 Existing Personalized and Risk-Adaptive Approaches

Personalized privacy, individual privacy accounting, and risk-adaptive protection describe different operations. Personalized privacy assigns an individual a prespecified budget ε_p, reflecting a stated preference or institutional policy. Individual accounting tracks or upper-bounds the privacy loss accumulated by each contribution, even under a shared mechanism. Risk-adaptive protection instead uses an empirical vulnerability estimate $\hat{V}_p$ to select the intervention applied to patient p. Song and Mittal's sample-specific scores illustrate a possible auditing input, whereas differential privacy defines the formal meaning of an assigned budget [15,41]. A patient's privacy preference, formally

assigned privacy budget, and empirically estimated membership vulnerability are distinct quantities and should not be treated as interchangeable.

Boenisch et al. provide a formal example of non-uniform assignment through Individualized DP-SGD [45]. IDP-SGD-Sample assigns different Poisson sampling probabilities: examples with larger budgets, and therefore weaker protection requirements, can be sampled more frequently. IDP-SGD-Scale changes per-example clipping bounds so that examples experience different effective signal-to-noise ratios while gradients remain efficiently aggregated. Both mechanisms connect predetermined budgets analytically to DP-SGD parameters and can improve utility relative to assigning every example the most restrictive budget. However, privacy levels are supplied before training rather than selected through membership-inference auditing. The formulation also assumes that each training point is contributed by a different individual. It therefore demonstrates individualized example-level guarantees, not protection for a clinical patient contributing multiple correlated records.

Moving from records to patients requires redefining adjacency and contribution control. Assigning ε_i to every record i in $\mathcal{R}_p$ does not automatically give patient p the same guarantee, because removing that patient changes all records in $\mathcal{R}_p$. One principled patient-centered construction would sample patients, aggregate their records into one contribution, clip the total contribution, and account for privacy under patient-level adjacency [3,41]. It must also address unequal record counts; otherwise heavily observed patients may contribute more gradient mass or exhaust budgets differently. Individualized DP-SGD could inform such a design, but its example-level analysis cannot simply be relabelled patient-level [45]. The protected unit, contribution bound, and composition rule must be specified before ε_p can be interpreted as a patient guarantee.

The missing step is a validated mapping from patient vulnerability to intervention, followed by independent re-auditing. Sample-specific scores [15] and the patient-linked clinical tail in Section 4 do not themselves implement that loop; Section 9 develops it as a proposed research framework.

Asiaee et al. provide a closer proof of concept for risk-guided intervention in differentially private synthetic data [46]. REPS spends part of the privacy budget to estimate record outlierness and then reduces the contribution weight of records estimated to be more exposed. Its benefits were clearest in a controlled simulation and smaller or inconsistent across three real tabular datasets, showing dependence on scorer, synthesis, and attack settings. The study is a 2026 preprint, operates at record level, and uses outlierness rather than calibrated patient-level MIA vulnerability; it neither handles multi-record patients nor validates a diagnostic model. Any data-dependent parameter selection must also be included in the privacy accounting or performed safely. REPS is therefore a mechanism-design proof of concept, not evidence that clinical risk-adaptive protection is established.

Adaptation can also create harms. Vulnerability estimates may be unstable across attacks or runs, causing inconsistent treatment of the same patient. Downweighting rare patients may reduce their influence and privacy loss while weakening learning for rare diseases or underrepresented populations. Protecting the current tail may expose a new vulnerable layer after retraining, consistent with the Privacy Onion Effect [35]. Small-group comparisons also carry statistical uncertainty [7]. Moreover, $\hat{V}_p$, ε_p, or the assigned protection category may itself reveal that a patient is unusual or has a sensitive condition. These quantities should be access-controlled and governed explicitly rather than treated as harmless metadata.

Existing personalized methods implement non-uniform budgets [45] or record outlier-guided synthetic generation [46]; neither validates an audit-to-intervention-to-independent-re-audit loop for multi-record clinical patients. A future adaptive policy must account for data-dependent selection in any formal DP claim and compare paired vulnerability and subgroup utility with a prespecified uniform baseline under Table 4.1.

7.6 Effects on High-Risk Patients and Subgroup Disparities

Benefits to the population, initially exposed patients and disadvantaged groups are distinct. This section records what defenses demonstrate; Section 8 formalizes paired benefit, residual vulnerability and disparity. The evidence tiers in Table 3.2 prevent general-benchmark defense results from being treated as clinical replication.

Knolle et al. provide the strongest direct clinical evidence. Strengthening record-level DP-Adam by lowering ε shifted the distribution of patient-linked membership vulnerability downward, showing that differential privacy can reduce empirical patient exposure rather than merely an aggregate attack statistic [3]. Nevertheless, the published analysis did not report paired patient-specific reductions for every privacy level, and extreme-tail subgroup composition was not re-evaluated at each level. At ε = 1, some multi-record patients had patient-level MIA AUC values above the upper bound implied for a one-record change. This does not constitute a violation of the record-level differential-privacy guarantee, because removing a patient changes several records; rather, it demonstrates a mismatch between the protected unit and the patient-level event. Thus, stronger record-level differential privacy reduced the patient-vulnerability distribution, but the study did not establish that initially highest-risk patients or subgroups obtained the largest absolute benefit.

Evidence that differential privacy narrows subgroup disparities is less direct. Kulynych et al. connected differential privacy theoretically to bounds on subgroup vulnerability and disparity; in non-clinical experiments, disparities became statistically undetectable in some settings, but with accuracy losses of approximately 5 to 45 percentage points depending on ε [7]. Truex et al. found greater vulnerability for minority classes in skewed data and showed that the effect of differential privacy depends on the budget, dataset complexity, and model [30]. Hu et al. found that tail-class vulnerability could persist under differential privacy even when long-tailed learning improved rare-class accuracy [31]. A common formal bound can constrain worst-case subgroup vulnerability, but it does not imply equal empirical vulnerability or equal absolute benefit, or elimination of the long-tail privacy gap. These findings are mainly non-clinical and class-level rather than patient-level studies of multi-record or intersectional clinical populations.

Per-example protection can be heterogeneous under general-benchmark defenses, including sample-specific scoring and distillation [15,44]. Regularization, augmentation and access controls have limited paired, patient-level clinical evidence [38,43].

Individualized DP-SGD assigns prespecified budgets at example level, while REPS uses record outlierness in synthetic-data generation; neither maps calibrated clinical patient vulnerability to a protected diagnostic model [45,46]. Rare-group utility and possible exposure of a new vulnerable layer after intervention remain open [35].

The clinical DP result described above does not establish preferential benefit for initially exposed patients or narrower subgroup gaps. Existing empirical and adaptive defenses require the paired comparisons formalized in Section 8.